

STEP CUE · PROBLEM STATEMENT

StepCue problem statement

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DOCUMENT CONTROL

Document ID	STC-PRB-001	Version	0.3
Status	Draft	Date	2026-09-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stepcue		
Source	docs/01-problem.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	One-insole question closed by Amish 2026-09-25 (STC-DDR-001); prevalence citation authors checked
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (problem, users, context, constraints, prior work with sources)
0.1	2026-09-24	Amish Chadha	Initial scaffold

StepCue problem statement

Freezing of gait is a sudden, brief inability to step that affects a large share of people with Parkinson's disease and is a major cause of their falls. Cueing (a rhythm the person can step to) often helps them get moving again, but the wearable devices that deliver a cue at the right moment cost several hundred pounds or more, and none is open for others to build, audit or adapt.

■ The problem

During a freeze the feet feel "glued to the floor" for a few seconds, most often when starting to walk, turning, passing through a doorway or approaching a target. A 2020 meta-analysis of 29 studies put the prevalence of freezing of gait at 39.9 % of people with Parkinson's disease, rising from about 22 % in the first five years after diagnosis to about 71 % after ten years (Ge et al., *Chinese Neurosurgical Journal*, 2020). Freezing is triggered by specific situations, which the research literature has catalogued in detail (*Frontiers in Neurology systematic review*, 2023), and it responds only partly to medication.

External cues are one of the best-supported management strategies: a rhythmic sound, a vibration or a line on the floor gives the brain an external timing signal for the next step (Ginis et al., *narrative review of cueing for freezing of gait*, 2018). Cueing that runs all the time can become irritating and less effective, so research systems detect the onset of a freeze and cue only then. A review of wearable cueing devices reports closed-loop research systems with detection sensitivities of about 73 % to 97 % and detection delays of about 0.5 to 3.2 s, using accelerometers on the shank, trunk or ankles (Sweeney et al., *Sensors*, 2019). Plantar pressure alone has also been used to predict freezes: one study using insole pressure data identified about 94 to 95 % of episodes with about 77 to 80 % sensitivity and 83 to 85 % specificity at the window level, about 0.8 s before onset, and found one insole about as good as two (Pardoel et al., *Frontiers in Neurology*, 2022; see also Shalin et al., *Journal of NeuroEngineering and Rehabilitation*, 2021).

Commercial options exist but are costly. The Path Finder laser shoe attachment is listed at £834 (*Parkinson's UK Tech Guide*). The CUE1+ chest-worn vibrotactile device is listed at £795 plus VAT, plus £60 a year for patches, and its sales were paused at the time of writing (*Parkinson's UK Tech Guide*). Neither is open hardware, and neither combines foot-pressure sensing with an on-demand cue at the foot.

The gap StepCue addresses: an open, low-cost, inspectable insole that senses how the foot is loading, recognizes the pattern of a freeze, and triggers a rhythmic cue only when it is needed, as a platform that researchers, clinics and patient groups can build and study.

■ Users and context

User	Need	Context
Person with Parkinson's who freezes	A cue that arrives quickly when a freeze starts and stops once walking resumes; nothing that changes how the shoe feels	At home and outdoors, in their own shoes, several hours a day
Care partner	Fewer falls; a simple device that does not need daily fiddling	Home
Physiotherapist or clinician	Adjustable cue type and tempo; a record of how often cues fired	Clinic sessions and home programs
Researcher	An open reference design and data format for studying detection and cueing	University labs, gait studies
Open hardware community	A reproducible, buildable design	Makerspaces

■ Constraints

- Garage-buildable prototype, about \$200 USD per unit of one instrumented insole and one heel pod (`budget_usd` in `project.yaml`), from off-the-shelf sensors and modules and 3D-printed parts.
- Fits in the wearer's own shoes: the insole must be no thicker than a normal replacement insole, and nothing rigid may sit under the heel or the metatarsal heads.
- Lithium cell and electronics stay outside the shoe, not under the foot.
- People with Parkinson's often have reduced hand dexterity: charging, fitting and switching must be possible with one hand and large features.
- Detection and cueing run on the device, with no phone or cloud needed for the core function; data stays with the wearer.
- Research and educational use only. StepCue is not a medical device and must not be relied on to prevent falls.

■ Out of scope

- Diagnosis, disease staging or any clinical decision.
- Stimulation beyond a perceptible cue (no electrical stimulation).
- Cloud services and remote monitoring.
- Fall detection or emergency alerts.

■ Open questions

- Which partner to co-design with first: a movement disorders clinic, a physiotherapy practice or a Parkinson's patient group? Proposed, awaiting Amish (no recommendation; STC-DDR-001, O1).
- One insole or a pair for the first build: decided by Amish, 2026-09-25, one insole on the side the wearer reports freezing most (STC-DDR-001, D1).
- Where to obtain labeled plantar-pressure freeze data to develop the detector before any human testing, which in any case needs ethics approval and belongs to TRL 4 or later.